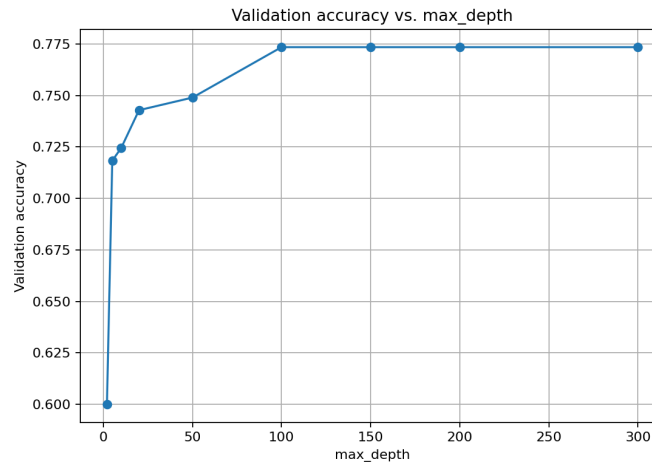


CP322 Assignment 1 - Q2 Decision Trees

Method. CountVectorizer converted 1968 real and 1298 fake headlines into word-count features. After deterministic shuffling (seed 42), the data were split into 2286 training, 490 validation, and 490 test examples. Decision trees used entropy/information gain.

Validation Results

Depth	2	5	10	20	50	100	150	200	300
Accuracy	60.00%	71.84%	72.45%	74.29%	74.90%	77.35%	77.35%	77.35%	77.35%



Selected model. The first maximum validation accuracy was **77.35%** at **max_depth=100**. Its actual depth was 100, and its held-out test accuracy was **75.51%**.

Part (c): Actual First Two Layers

```
|--- donald <= 0.50
|   |--- trumps <= 0.50
|   |   |--- truncated branch of depth 99
|   |   |--- trumps > 0.50
|   |   |   |--- truncated branch of depth 4
|--- donald > 0.50
|   |--- the <= 0.50
|   |   |--- truncated branch of depth 34
|   |   |--- the > 0.50
|   |       |--- truncated branch of depth 25
```

The root split uses the word feature shown on the first line above. The two indented branches show the next decision layer. These values were extracted directly from the selected depth-100 model trained on the supplied real.txt and fake.txt files.